

surveydown

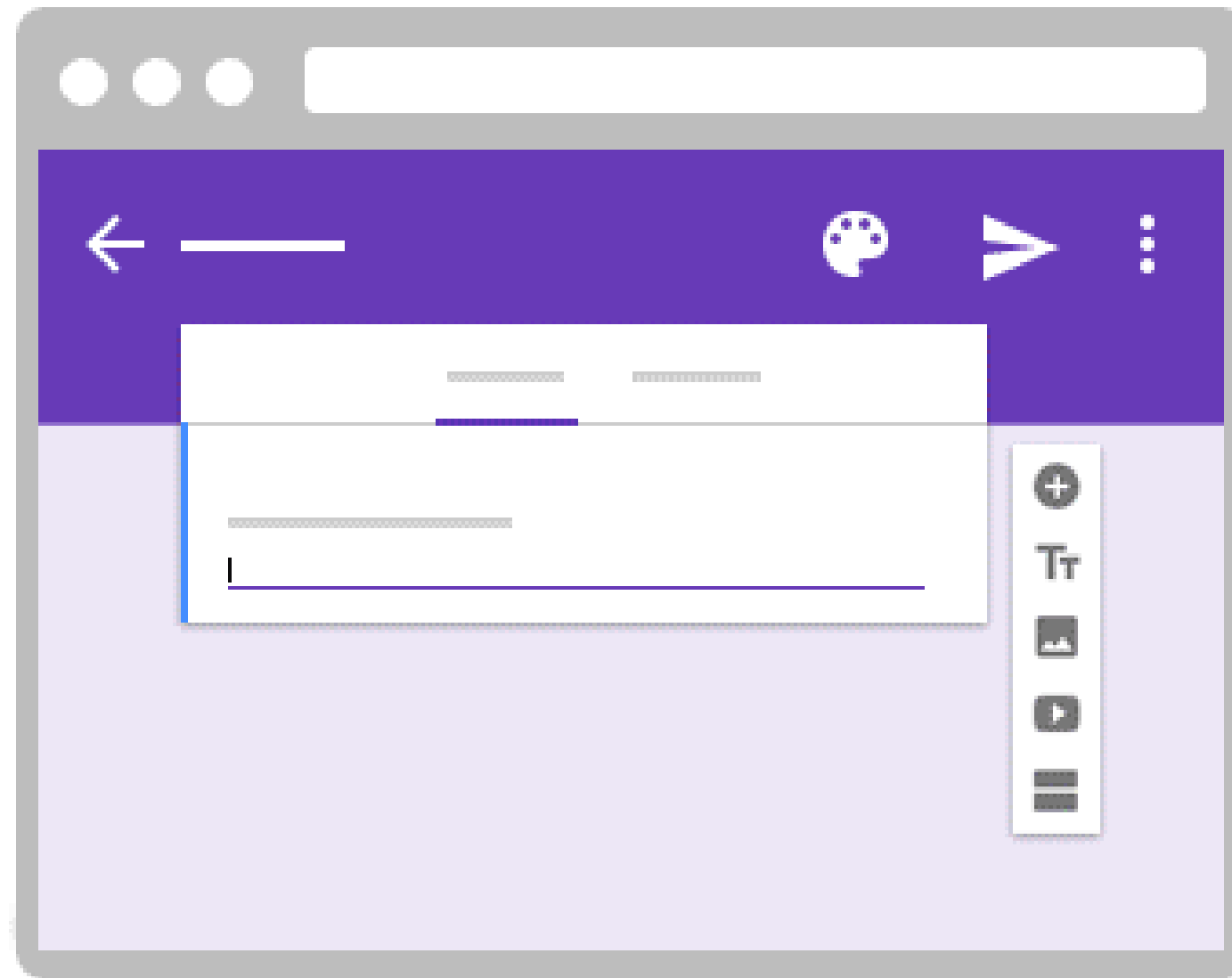
An Open-Source, Markdown-Based Platform
for Interactive and Reproducible Surveys

Pingfan Hu, Bogdan Bunea, John Paul Helveston
George Washington University

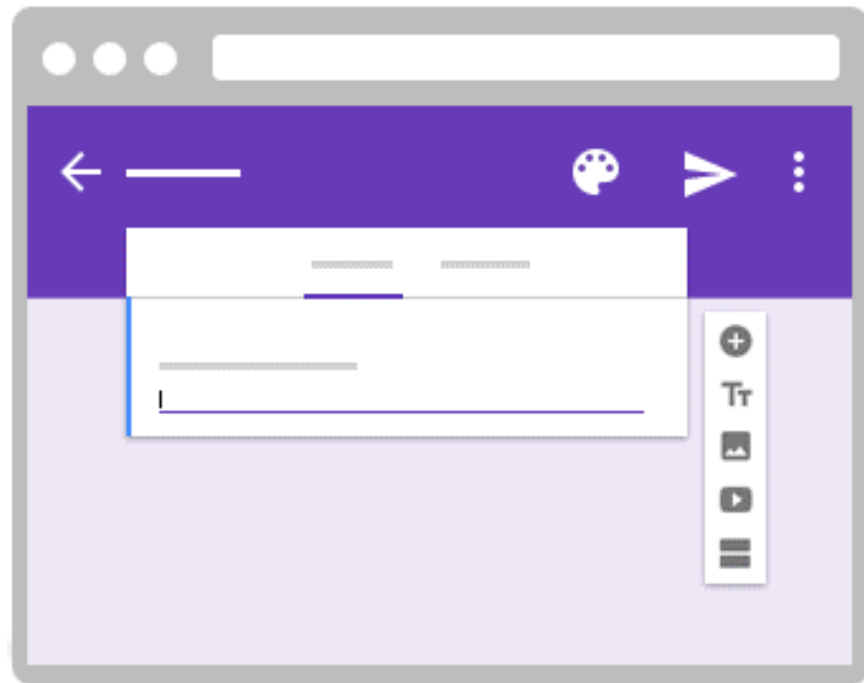
GW OSCON 2025



Typical experience making a survey



Typical experience making a survey



✗ Reproducible

✗ Data Control

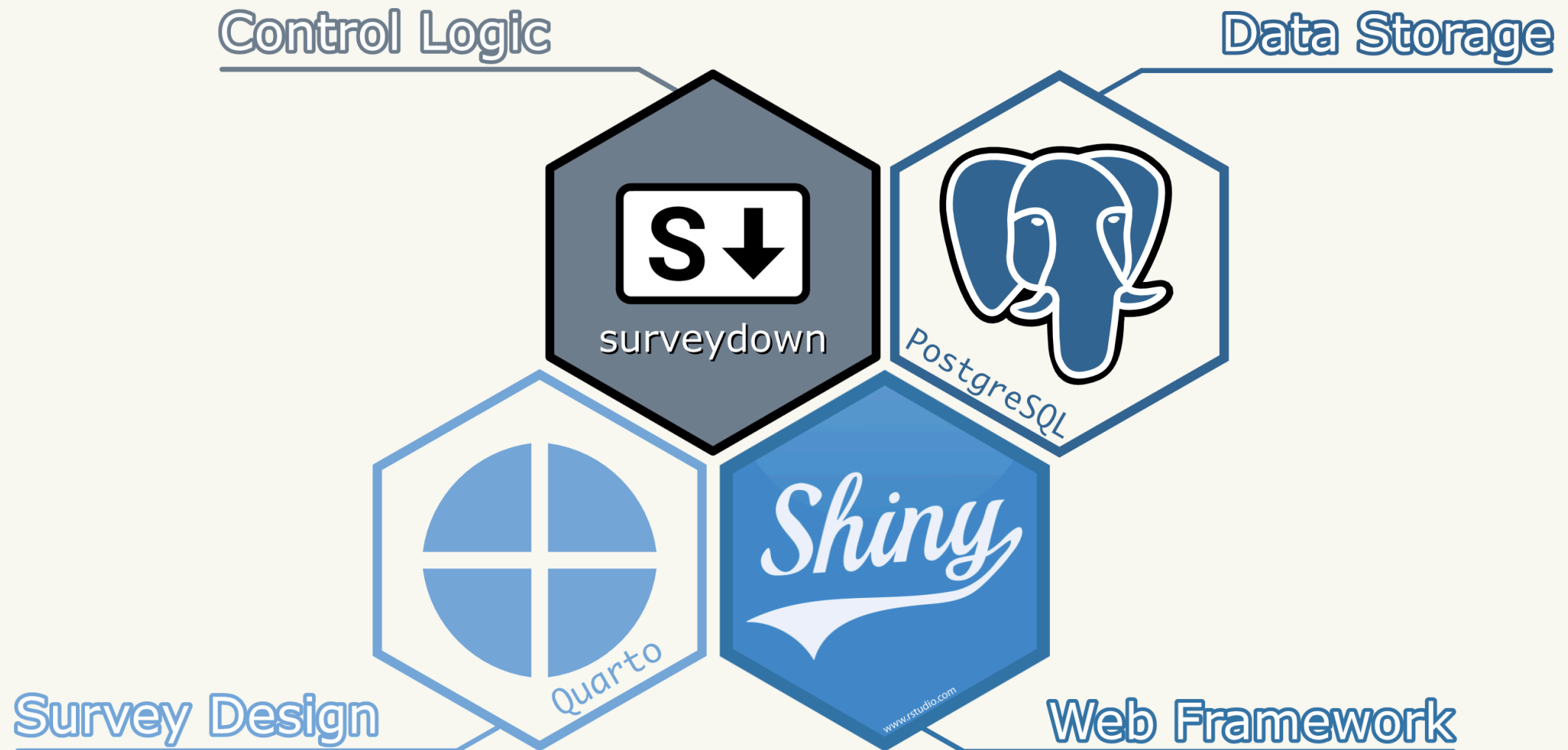
⚠ Free to Use

⚠ Open Source

⚠ Easy Collaboration

⚠ Feature Packed

Introducing surveydown



Components

`survey.qmd`

A **Quarto doc** defining the main survey content (pages, text, images, questions, etc.).

`app.R`

An **R script** defining the survey Shiny app. It sets up the database and server, and launches the survey.

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to our survey!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id = "penguins",
19   label = "What's your favorite penguin?",
20   option = c(
21     "Adélie" = "adelie",
22     "Chinstrap" = "chinstrap",
23     "Gentoo" = "gentoo"
24   )
25 )
26
27 sd_next()
28 ```
29
30 --- end
31
32 This is the last page of the survey.
33
34 ```{r}
35 sd_close()
36 ```
```

survey.qmd

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YAML Header

Load Package

survey.qmd

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Define pages using triple-dash markers:

--- page_id

Page 1 (Welcome)

Page 2 (End)

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Contents of Page 1

survey.qmd

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Question Definition

app.R

```
1 # Load Package
2 library(surveydown)
3
4 # Database Credentials
5 sd_db_config()
6
7 # Connect to Database
8 db ← sd_db_connect()
9
10 # Server Setup
11 server ← function(input, output, session) {
12
13   # Skip logic
14   sd_skip_forward(...)
15
16   # Conditional display
17   sd_show_if(...)
18
19   # Server settings
20   sd_server(
21     db = db,
22     ...
23   )
24 }
25
26 # Launch Survey
27 shiny::shinyApp(ui = sd_ui(), server = server)
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```

Database Setup

We recommend using **Supabase** or any **PostgreSQL** database for the database setup.

app.R

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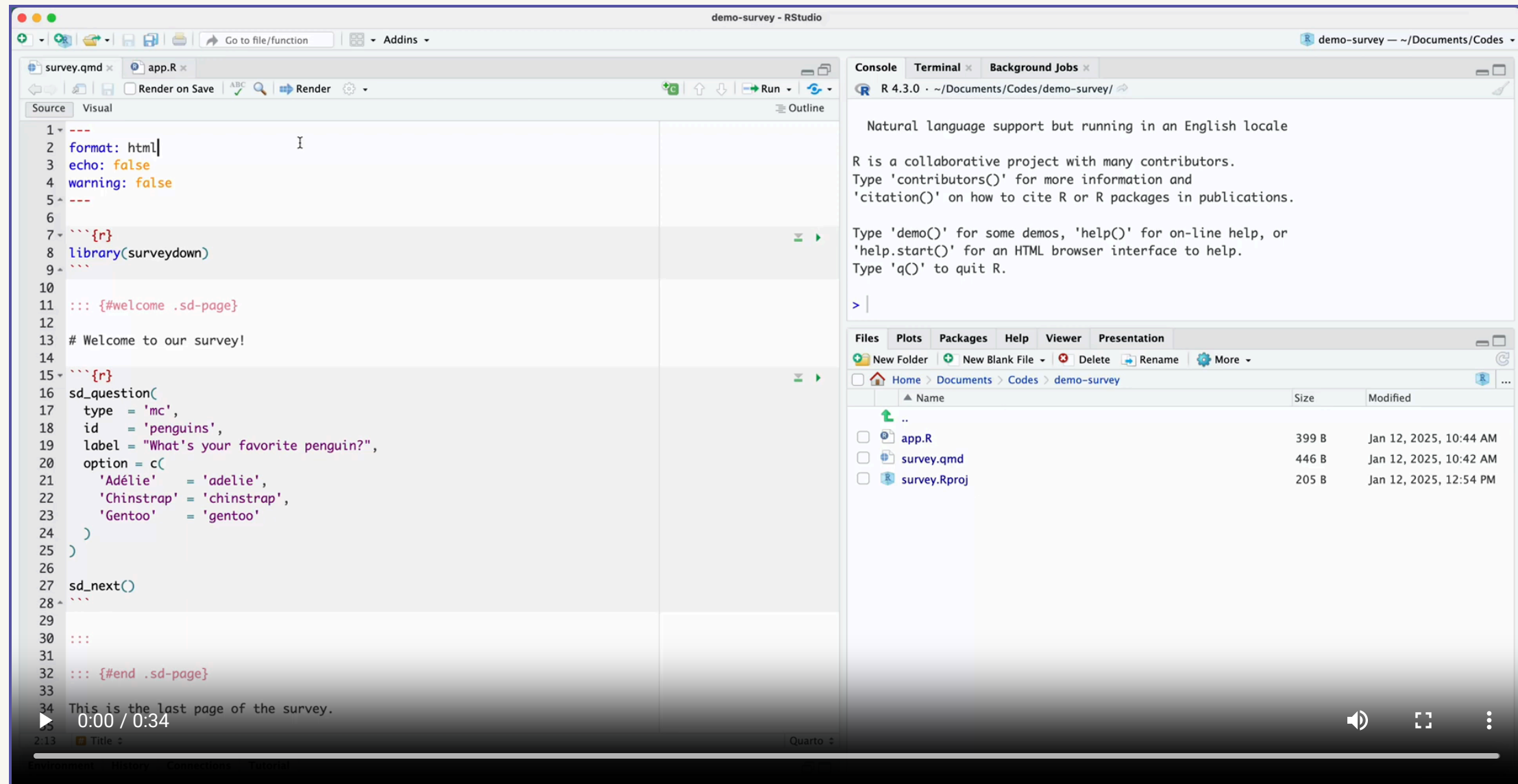
Server Setup

app.R

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```

Launch Survey

Locally running the survey

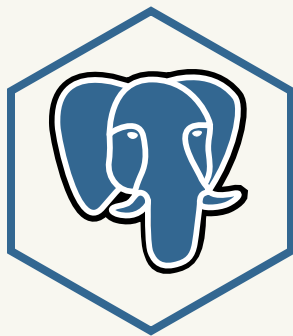


Data saved locally in CSV for previewing purposes.

Online Deployment

Data Storage

- Any PostgreSQL database
- Recommend **Supabase** (Free)
- Use `sd_db_config()` to set up



PostgreSQL



Supabase

Deploy

- Recommend **shinyapps.io** (Free):

```
rsconnect::deployApp(appName = "my_survey")
```

- Other Platforms:

- Posit Connect Cloud
- Hugging Face
- Heroku

Online Data Storage + Online Survey URL

Local Dashboard App

The screenshot shows the RStudio interface for a project named 'question-types'. The editor window displays an R script with the following content:

```
1 # Package setup -----
2
3 # Install required packages:
4 # install.packages("pak")
5 # pak::pak('surveydown-dev/surveydown') # Development version from github
6
7 # Load packages
8 library(surveydown)
9
10 # Database setup -----
11 #
12 # Details at: https://surveydown.org/manuals/storing-data
13 #
14 # surveydown stores data on any PostgreSQL database. We recommend
15 # https://supabase.com/ for a free and easy to use service.
16 #
17 # Once you have your database ready, run the following function to store your
18 # database configuration parameters in a local .env file:
19 #
20 # sd_db_config()
21 #
22 # Once your parameters are stored, you are ready to connect to your database.
23 # For this demo, we set ignore = TRUE in the following code, which will ignore
24 # the connection settings and won't attempt to connect to the database. This is
25 # helpful if you don't want to record testing data in the database table while
26 # doing local testing. Once you're ready to collect survey responses, set
27 # ignore = FALSE or just delete this argument.
28
29 db <- sd_db_connect()
30
31 # Server setup -----
32
```

The console window shows the R version 4.4.2 and the current directory: `~/Documents/GitHub/surveydown/demos/question-types/`. The file explorer on the right shows the project structure:

Name	Size	Modified
..		
app.R	1.7 KB	Mar 20, 2025, 10:08 PM
survey.Rproj	414 B	Mar 21, 2025, 4:21 PM
preview_data.csv	1.8 KB	Mar 20, 2025, 1:39 PM
.gitignore	5 B	Mar 19, 2025, 9:26 AM
survey.qmd	9.6 KB	Jan 22, 2025, 8:34 AM
_survey		
rsconnect		

Templates and demos are provided on surveydown.org
for an easy start.

Summary of Features

Reactivity and randomization

Dashboard for survey data

Cookies for restoration

URL parameters for redirection

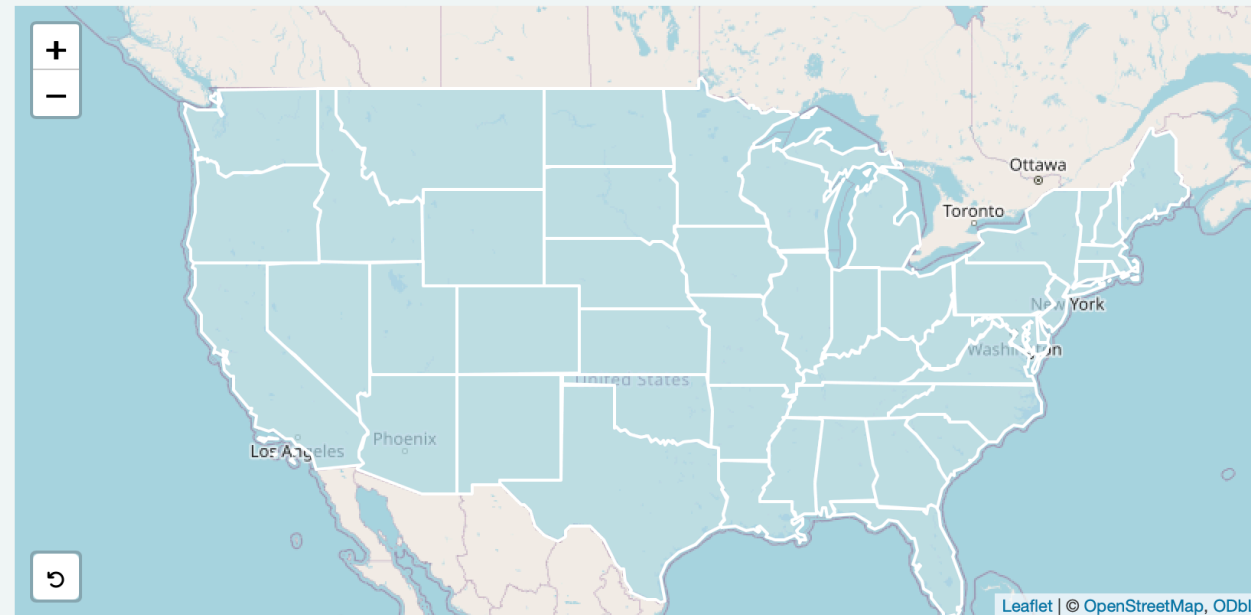
Ability to start from a selected page

Custom interactive questions

Example of a custom interactive question

Demo - Custom Leaflet Map

Click on the state you live in:



You live in:

Next

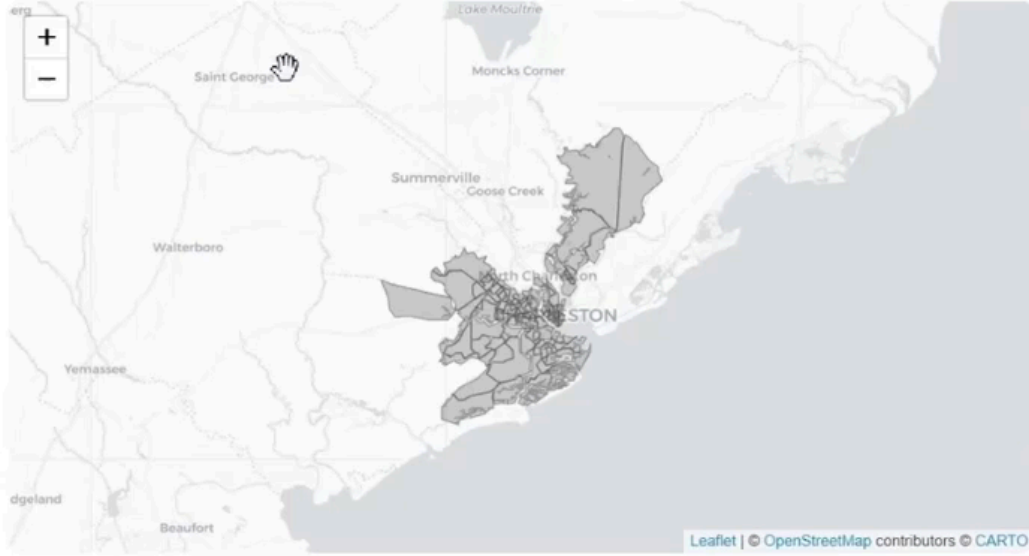
Made with [surveydown](#)

 [Source Code](#)

Map Question Showcase

Question 1: Social Vulnerability is how likely people are to face greater harm or impact on their lives after disruptive hazard events like flooding based on social or economic conditions like income or transportation access.

Please review the map of the City of Charleston, South Carolina, and select the areas you believe are more socially vulnerable to the impacts of hazardous events.



The map displays the City of Charleston, South Carolina, with various neighborhoods labeled. The map is interactive, showing a hand cursor over Saint George. The map is overlaid with a grid of colored polygons representing different levels of social vulnerability. The map is sourced from Leaflet, OpenStreetMap contributors, and CARTO.

Next Question

0:00 / 0:56

Source: Dr. Selena Hinojos, GWU

Future Work

GUI for defining surveys

More question types

...and more!

You can contribute!

Check out surveydown.org for more!